pyAmpliCol at one loop

Implementation and live performance report

2026-10-18 · revision fa599885 · AMD EPYC 9754 · one pinned CPU thread · shared host

First development snapshot. The numbers below evaluate complete Born–loop interference for the listed internal QCD controls: all physical helicities, full SU(3), UV counterterms, model-supplied R_2, and all three Laurent coefficients. Native batched arithmetic is driven by the current Python coordinator. These are measured implementation diagnostics; the matched MadLoop/Recola/OpenLoops speed gates have not been run.

Complete evaluation in binary64

Each cell is **microseconds per phase-space point**, the median of five warm batch calls. The batch repeats three frozen, valid control points. It does not reuse computed event results. Exact decimal ingress and Python batch scheduling are inside the timer; output formatting and reference comparisons are outside.

Table 1: Warm complete runtime. B is the number of physical phase-space points in one call. All rows use model-rule R_2.

Process	$B = 1$	$B = 8$	$B = 32$	$B = 128$
$u\bar{u} \to e^- e^+$	583.7	170.0	142.0	147.4
$u\bar{u} \to d\bar{d}$	1733.4	814.9	719.4	764.0
$u\bar{u} \to t\bar{t}$	2006.2	940.0	811.9	745.0
$u\bar{u} \to gg$	6834.6	4701.6	4125.2	3099.8
$gg \to gg$	33655.7	19119.7	15208.4	14280.3

Table 2: Cold construction and first evaluation, measured separately. One cold observation per process; no statistical uncertainty is inferred.

Process	Generation / s	First $B = 3$ / ms	Peak RSS / MiB
$u\bar{u} \to e^- e^+$	2.43	21.6	132
$u\bar{u} \to d\bar{d}$	7.09	63.0	180
$u\bar{u} \to t\bar{t}$	6.39	90.6	183
$u\bar{u} \to gg$	21.16	616.2	321
$gg \to gg$	98.74	11531.7	1085

Generation includes source UFO capture/lowering, graphs, colour and recurrence preparation, counterterm matching, and creation of JIT/reducer plans. Imports, dependency installation and the first evaluation are excluded. Lazy first-call work is therefore visible separately. Peak RSS is the process high-water mark across construction and all batch tests, not an event-buffer estimate.

Development build: Rust dev profile, local module opt3 / dependencies opt2, no LTO; dev assertions and overflow checks retained. Community opt2; SymJIT. This is not a frozen release-performance build. One pinned core (CPU 382), one thread, shared AMD EPYC 9754 host. Five calls describe local variation only. Source capture and cold plans are rebuilt per process. "—" means not measured in this snapshot. Real-mass EW, higher multiplicity and direct-R_2 timings remain to be added under this same protocol.

MEASUREMENTS · COMPONENT BOUNDARIES

Where the event time goes

The requested decomposition separates numerator work, integrand reduction and scalar master integrals. A complete event also needs states, kinematics, Born weighting, UV/R_2 currents and output assembly. Those costs must remain visible; the three core columns need not sum to the end-to-end time.

Table 3: pyAmpliCol at $B = 32$, in µs per point. Stage and "instr." columns are medians of five instrumented calls; "total" repeats the independent uninstrumented baseline. Component medians need not sum to the median instrumented total.

Process	Numerator	NinjaRS reduction	Scalar masters	Other	Instr. total	Complete total
$u\bar{u} \to e^-e^+$	24.7	0.9	0.3	71.7	96.8	142.0
$u\bar{u} \to d\bar{d}$	294.0	7.8	9.1	381.0	691.4	719.4
$u\bar{u} \to t\bar{t}$	334.9	16.1	18.3	520.8	889.8	811.9
$u\bar{u} \to gg$	1778.4	21.5	17.7	1388.7	3195.9	4125.2
$gg \to gg$	9185.4	88.4	68.8	4515.4	14166.8	15208.4

Table 4: Fresh reference performance at $B = 32$, in µs per point, on the same control stream and pinned CPU. Full physical coefficients agree. Input and stability policies differ as described below; these are development comparisons.

Process	MadLoop numerator	MadLoop reduction + masters	MadLoop total	Recola total	OpenLoops total
$u\bar{u} \to e^-e^+$	6.5	35.9	46.8	—	—
$u\bar{u} \to d\bar{d}$	55.8	197.1	255.6	2293.7	543.0
$u\bar{u} \to t\bar{t}$	94.5	256.9	318.7	2953.5	675.9
$u\bar{u} \to gg$	205.5	127.3	404.4	4884.5	1485.2
$gg \to gg$	1915.9	824.8	2283.3	32811.3	10357.6

Timer definitions

Numerator is loop coefficient DAG execution plus reducer numerator callbacks and coefficient-head preparation. **Reduction** excludes callbacks and scalar integrals. **Masters** is the actual oneloopmaster adapter time, with its existing cache policy. **Other** contains states, exact kinematics, Born and UV/R_2 work, weighting, assembly, transport and clock overhead. Instrumented components and uninstrumented runtime are separate observations; neither is rescaled to force agreement. Raw five-sample variation is retained with the report.

MadLoop coefficient construction excludes shared wavefunctions supplied by HELAS; its loop-CT call window includes reducer interfaces, masters and retries. Recola/OpenLoops totals use native public calls in compiled loops. Provider input parsing is excluded; pyAmpliCol includes exact decimal ingress and Python batch scheduling. Provider rescue stays enabled where supplied; pyAmpliCol has no rescue here. Instrumented stage clocks are separate from original-binary totals. The current complete totals are slower than MadLoop in every listed case.

Acceptance is still open. On each designated high-multiplicity process, pyAmpliCol must be faster than MadLoop and at most 1.30 times each available matched Recola and OpenLoops result, for warm single-point latency and native batch throughput. The faster complete, validated SM R_2 strategy may be used. A slow process cannot be hidden by a geometric mean. These measurements do not establish performance acceptance.

How this report will evolve

Replace missing cells only with matched measurements. Keep generation, first evaluation and warmed runtime separate. Add distinct frozen event streams, paired provider timing blocks, rescue rates, uncertainty and memory before evaluating a performance gate. Keep historical diagnostics tied to their own revision instead of merging them into the current table.

ARCHITECTURE · PHYSICAL CONTRACT

The quantity being computed

The new engine targets one-loop amplitudes interfered with a Born amplitude. It does not compute the squared one-loop amplitude and does not admit loop-induced production. A Born zero at an individual point or helicity remains a valid event of a supported process.

For Born order a, loop order b and squared coupling order $s = a + b$, the ordinary uncorrelated observable is

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\Re\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0, -1, -2\}.$$

The rectangular matrix G_{BL} connects the actual Born and loop colour bases. The normalization $\mathcal{N}$ includes declared initial averages and final identical-particle factors. Couplings are included; the result is neither divided by the Born value nor normalized by an extracted strong coupling.

The returned coefficient order is **(finite, single pole, double pole)** in the MadLoop convention,

$$V = C_\varepsilon \left(V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2} \right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}.$$

The finite conversion used in the matched Recola controls is $V_0^{\mathrm{ML}} = V_0^{\mathrm{Recola}} + \left(\frac{\pi^2}{6}\right) V_{-2}$; both poles are unchanged. Such conversions act on the Laurent series, not on a process-dependent fitted factor. Scheme, scale and finite renormalization conventions are retained separately.

Coupling selection is a graph property

`loop_coupling_orders` requests perturbative corrections. Eligibility depends on the actual loop interactions and internal species, followed by explicit Born–loop pairings. An order difference alone cannot identify the correct graphs. In the mixed four-quark control,

$$B = B_{20} + B_{02}, \quad L = L_{40} + L_{22} + L_{04},$$

the squared order $(4, 2)$ receives both $B_{20}^* L_{22}$ and $B_{02}^* L_{40}$. Their identities survive through counterterm matching and output assembly. Joint QCD/QED selection also admits mixed graphs absent from either singleton selection. Independent MadLoop graph controls give 16 QCD, 15 QED and 35 joint loops for the retained mixed model.

What the measured processes include

Control	Initial / identical normalization	Physical completion
$u\bar{u} \to e^- e^+$	$\frac{1}{36}$	Photon Born; QCD virtual
$u\bar{u} \to d\bar{d}$	$\frac{1}{36}$	Local UV and R_2
$u\bar{u} \to t\bar{t}$	$\frac{1}{36}$	Local and external-field UV; R_2
$u\bar{u} \to gg$	$\frac{1}{72}$	Local/field UV; R_2
$gg \to gg$	$\frac{1}{512}$	Local/field UV; R_2

The QCD scattering controls use $\alpha_s = 0.118$, $\mu_R = 1000$ GeV, stable $m_t = 173$ GeV, five renormalization flavours and six quarks in loops; the other quark masses vanish under explicit control restrictions. Drell–Yan uses $\alpha^{-1} = 132.507$. Initial spin and colour are averaged, and physical external helicities are summed. The two final-gluon controls include $\frac{1}{2}$.

ARCHITECTURE · MODEL AND COUNTERTERMS

Preserve the model before lowering it

Loop models contain information that a tree-only loader can discard: Laurent-valued parameters and couplings, UV and R_2 categories, loop-particle groups, external-field factors and duplicate vertex names. The new raw model retains positional declaration identities and these fields before constructing the exact scalar and covariant-kernel representations.

Raw source
Exact declarations
Species and loop groups

→

Prepared model
Scalar programs
Colour × Lorentz entries

→

Process plan
Admitted graphs
Matched UV / R_2 / fields

Exactness begins at input

Statically identifiable source literals are preserved as text, including long decimal and Laurent coefficients. String-valued expressions remain exact. Arbitrary Python computations or mutations do not gain an exactness claim merely because their imported result is a number. Explicit exact runtime overrides and fixed parameters flow through the same retained scalar owner.

Symbolica performs substitution, series extraction and expression preparation. Source helper functions and their branch semantics are retained. Each selected scalar has a value and an active diagnostic expression; native evaluation checks both. A nonfinite or unsupported active branch cannot silently inherit a successful amplitude status. Per-point renormalization scales reach the same scalar program used by counterterms.

Tensor structure and physical frame

Each basis entry retains its scalar coefficient, native Lorentz tensor, native colour tensor and ordered particle ports. Multi-colour quartic vertices remain sums of local terms. There is no reconstruction of tensor algebra in pyAmpliCol: Idenso/Spenso/FeynKit own contractions, traces, component libraries and index transformations.

Source spinor slots and the native physical wavefunction convention require an explicit, bounded frame map. The FFV map renames typed external Dirac ports simultaneously. The newly admitted bounded FFS map also exchanges `ProjP`/`ProjM`, with 580 independent ALOHA controls; the FFV map and propagators remain unchanged. Neither map rewrites momentum labels or supplies a universal gamma-five transpose. Unvalidated source structures retain an explicit boundary.

Counterterms follow admitted loops

UV-tree terms, local loop counterterms and external-field factors have different eligibility rules. A standalone counterterm tree is insufficient evidence that its loop correction belongs to the request. Local R_2/UV matching retains the actual attachment currents, ordered binding legs, local coupling order and allowed loop-species alternatives. Wildcards and repeated alternatives preserve their source semantics and multiplicity. Fresh native tree graphs provide the insertion signs and propagators.

External-field factors multiply the applicable Born currents with each physical leg retained. Local and field UV remain separately inspectable, then join the same ordered interference. Model-rule R_2 currents already contain their loop normalization; the assembly does not multiply them by a second loop measure.

Evidence boundary. Complete bounded QCD controls and real-mass neutral-current charged-lepton EW components are validated. That does not certify every rule in a supplied UFO, arbitrary restriction cards, all gauges, or general BSM renormalization.

ARCHITECTURE · RECURRENCE AND FAMILIES

Local tensors, reusable coefficient currents

The production direction is a dedicated loop recurrence. FeynKit supplies the graphs, directed incidence, loop signs, symmetry factors, cuts and affine momentum routing. pyAmpliCol turns those local ingredients into a coefficient dataflow graph and schedules its numerical evaluation.

A current retains the coefficients of a symmetric polynomial in the physical loop momentum. For a local tensor T and previous current J, the next step is the contraction $T \cdot J$, followed by coefficient extraction using Symbolica. Only the local transition is prepared symbolically. A whole-diagram symbolic trace is used in small independent oracles, not as the numerical recurrence.

Coefficient dataflow. External states and tree attachments feed local transitions; loop closure produces ordered coefficient outputs. Native saved evaluator instructions update these outputs in batches. Each original graph retains its colour terms, coupling sector and routing map even when it shares an identical numerical schedule with another graph.

Topology remains attached to the numerator

Reduction needs a denominator set as well as a polynomial. Each graph carries its routed propagators and a dependency-proved map into an integral family. Numerators are transported through that map before combination. Cyclic or reversed order is not inferred from a loose list of masses: momenta, incidence, external attachments and the actual transformation must agree.

The family plan retains bridges and local tree denominators. Native graph factors carry fermion and ghost signs. Multi-term local vertices are expanded into ordered recurrence outputs, while the original graph/family identities remain intact. This separates a numerical reuse opportunity from a physical multiplicity or symmetry factor.

Contract before reducing when linearity permits

For an interference observable, Born amplitudes and the rectangular colour weights can multiply numerator coefficients before integration. Compatible graphs, helicities and colour terms then share a reduction within their family and coupling sector. The reducer sees the coefficient combination needed by the observable, rather than every independent loop amplitude.

Complex weights are retained until the correctly directed bilinear is formed. Taking a real part too early loses information needed by colour histories and other off-diagonal observables. Counterterm interference uses the same Born currents and colour conventions, with its own local scalar factors.

Sharing that is safe under parameter updates

Compiled kernel reuse keys the saved instructions, settings, carrier precision and output layout. Evaluators may share immutable compiled code while owning independent work buffers. Whole-schedule numerical reuse additionally requires identical ordered inputs under an exact model-binding proof. Equal floating values at one point are insufficient.

The current process cache shares compatible Born, loop and counterterm kernels; exact whole-schedule groups cover Born and loop execution. It does not cache event results between calls. Cold plans independent of arithmetic, persistent artifacts and broader subtree reuse remain active work.

Current NinjaRS support bounds are enforced explicitly. The generic all-precision path is limited to six loop propagators and validated weighted rank through $n + 1$; unsupported ranks are rejected rather than truncated. Required seven-propagator coverage remains a dependency-owner gate.

ARCHITECTURE · COLOUR AND OBSERVABLES

Full colour without dense event tensors

The current complete QCD path contracts colour at plan construction using native fixed-SU(3) tensors. It forms the properly scoped bra–ket expression, uses dependency colour simplification and caches the exact pair. The numerical event path receives the resulting rectangular weights. Dense SU(3) components remain an independent small-control oracle, not the production colour sum.

Closed residual contractions use the dependency's existing projector expansion, trace/chain operations and canonicalization. Minimal upstream fixes address specific demonstrated tensor-lowering or invariant-contraction defects. No private group identities or numerical colour fitting are inserted into the application.

LC and NLC are structural projections

For a certified formal-N_c model, all contributions belonging to one squared order and operator family are combined before finding the common highest power P. With N_f held fixed,

$$\mathrm{LC} : N_c^P, \quad \mathrm{NLC} : N_c^P, N_c^{P-1}, N_c^{P-2}.$$

Odd suppression powers are retained. The same P spans the relevant helicities, epsilon sectors and source contributions; choosing a different leading power for each term would change the approximation. Exact native algebra handles Laurent coefficients and semantic C_A, C_F, T_R relations.

Complete bounded quark-process colour projections have controls. The active `loop_sm` four-gluon R_2 model contains fixed-$N_c = 3$ coefficients. Its full-SU(3) validity does not imply a formal continuation, and numerical constants are never promoted heuristically to N_c. Validated direct completion can provide a different continuation only within its certified domain.

Directed colour and spin sources

Correlated virtual observables require both rectangular cross terms. For operator histories A and B, their generic form is

$$B_A^\dagger G L_B + L_A^\dagger G B_B.$$

For unequal histories this is not generally $2\mathfrak{R}\left(B_A^\dagger G L_B\right)$; the result can be complex. Existing colour-connection metadata and native charge tensors provide the operator action and signs. Current internal controls compose bounded histories with Born, bare loop, local counterterms and field factors.

Literal complex spin-source vectors enter the admitted physical external ports without an implicit conjugation, normalization or projection. Spectator states are summed and a replaced leg is counted once. The source/current frame and crossing factors remain explicit. Scalar and massive-vector layouts reuse native wavefunctions; a generic source API is not yet certified by these controls.

Flavour mapping and FFT remain distinct work

An exact species-symmetry certificate now compares the entire retained ordinary, UV, R_2 and external-field plan, including prepared native structures and parameter bindings. Independent massless d and s catalogues pass; unequal masses, Yukawas or UV logarithms reject. This is a prerequisite for mapping, not yet external-leg, momentum, spin-source or colour-history transport.

Optional FFT contraction is an agreed feature but is not part of the measured path. Its cost model, exact observable reconstruction and composition with the other features remain acceptance gates.

ARCHITECTURE · NATIVE BATCHES AND PRECISION

One numerical path, three real carriers

The internal numerical core uses binary64, DoubleFloat and arbitrary-precision carriers throughout coefficient evaluation, kinematics, NinjaRS reduction, scalar masters, counterterms and contractions. It does not compute a binary64 answer and promote it afterward. Fixed-carrier precision metadata follows the actual numerical types; arbitrary precision carries its requested precision.

Exact decimal or rational input is parsed before carrier conversion. S-matrix and admitted on-shell invariants are constructed through the exact input boundary; source mass/coupling formulas are evaluated natively in the chosen carrier. Controls with differences below binary64 resolution verify that DoubleFloat and arbitrary precision retain the input information through the output. Source scalar callbacks and their diagnostics use those same carriers.

Batching is an execution contract

Points, helicities and coefficient components have explicit ordered layouts. Native buffers retain original point identities through compaction, tail batches, failure masks and result scatter. Changing sizes, empty batches and repeated calls are covered. Python coordinates batch plans; it does not evaluate each phase-space point in an interpreted numerical loop.

The coefficient scheduler uses bounded row tiles and reusable scratch storage. Its present 16-row default is a measured development choice, not a universal optimum. Dependency evaluators own their arithmetic workspaces. Binary64 can use SymJIT; high-precision paths remain native batches without an unsupported SIMD claim. Public standalone SDK integration is a separate unfinished layer.

Reducer and master interfaces

NinjaRS receives the polynomial layout, routed denominator data, exact kinematic invariants and carrier-appropriate inputs. The adapter retains weighted rank: μ^2 counts as two powers of loop momentum. Separate explicit-epsilon sectors reduce independently and combine by Laurent multiplication. A newer dense-μ numerator adapter is available as an additive internal option; the headline timings specify the actual `fast` engine they use.

The native module and Community producer use the same underlying numerical and symbolic owners. Python distribution and Rust kernel version labels are recorded separately. A genuine isolated development-package install now passes a public ordinary-tree generation/evaluation smoke. Publishable dependency packaging, loop artifacts and cross-language virtual SDK delivery remain open.

Success, accuracy and rescue are different facts

A reducer's success flag is not an accuracy certificate. Native checks reject nonfinite input, state, coefficient and counterterm values, preserve the original failed point identity, and permit an explicit partial-result contract. Optional cross-precision checks compare complete outputs and can rescue selected failed or disagreeing points at higher precision without changing R_2 strategy.

The result records requested/evaluated precision, strategy, status and available agreement evidence. Failed points cannot silently appear as zero or be declared accurate. Existing bounded rescue controls are not a universal stability policy: threshold choices, complete feature composition and adversarial tests remain open. None of the headline runtime cells enables rescue or cross-precision verification, and each is labelled accordingly.

Known numerical gate. An equal-value, differently tracked arbitrary-precision mass construction exposed a scalar-box branch discrepancy in a CMS zero-width control. This is retained as a dependency-owner accuracy issue. Extra digits alone are not evidence that a branch-sensitive result is correct.

ARCHITECTURE · RATIONAL TERMS

Two complete strategies, one observable

The model strategy combines the four-dimensional numerator contribution with the model's effective R_2 rules and UV terms. The direct strategy instead computes the required evanescent numerator remainder and retains the same UV completion. The strategy is explicit in the result; a caller cannot relabel an executed model result as direct.

Four-dimensional reduction already produces denominator-related rational terms. The presence of μ^2 in a reducer interface therefore does not, on its own, prove that numerator-origin R_2 is complete. The direct extension must account for extra-dimensional momentum, metric/spin traces and explicit epsilon factors.

Local evanescent state, not a whole-loop trace engine

For a certified closed vectorlike fermion cycle, the local component construction uses dependency gamma matrices with one auxiliary evanescent slash. Odd auxiliary powers survive every open transition; only the closed trace selects even parity. Independent full-D local-graph oracles check routed bubbles, triangles and boxes. The auxiliary gamma-five realization is not a general physical chiral or dimensional gamma-five prescription.

For the admitted massless Yang–Mills cycles, dependency metric and momentum rules are continued to an explicit physical/aligned/spectator state space. The remaining transverse-state multiplicity is retained symbolically in $D = 4 - 2\varepsilon$. Local cubic and quartic colour terms, closure, routing and spectator weighting are checked independently. Ghost cycles keep their native statistics and their ordinary bare contribution.

Epsilon sectors matter after integration

If the numerator is $N_0 + \varepsilon N_1 + \varepsilon^2 N_2$ and each reduced sector has finite/single/double coefficients $\left(F_j, S_j, D_j\right)$, the assembled result is

$$V_0 = F_0 + S_1 + D_2, \quad V_{-1} = S_0 + D_1, \quad V_{-2} = D_0.$$

Dropping explicit epsilon terms can therefore change the finite answer even when the mu-dependent terms are retained. Native reduction controls exercise this convolution in all three carriers. Source templates, full-graph algebra and integrated physical results have separate checks.

What is supported internally today

The complete retained full-colour $gg \to gg$ control admits the local direct kernels, includes UV completion, and agrees with model-rule components and independent reference results in all three arithmetic modes. Bounded cross-precision verification and direct-strategy rescue are also implemented. Public direct dispatch remains behind its integration gates.

Open fermion chains, general mixed-spin loops, massive gauge/Goldstone sectors, physical axial or anomaly-sensitive structures and arbitrary BSM dimensional extensions are not inferred from that success. Unsupported direct domains need compatible model-rule completion. No universal massless FDH-to-HV shift is assumed for massive EW or general models.

Selection and performance. Automatic policy may prefer a certified direct strategy, but completeness is mandatory. Model and direct costs must be measured as standalone complete evaluations. The present headline table uses model rules throughout; a new direct-versus-model comparison is still due after recent colour and scheduling changes.

VALIDATION · NEXT GATES

What this snapshot establishes

The measured controls are useful working vertical slices: model import, graphs, local recurrence, full colour, native states, reduction, UV/R_2 and ordered physical output all run together. Independent reference comparisons separate bare, local UV, external-field UV and R_2 so cancellations in the complete answer cannot conceal a wrong component.

Area	Current boundary
Complete QCD	Internal DY, four-quark, top-pair, quark/gluon and gluon controls; all three carriers.
Real-mass EW	Source-owned neutral-current charged-lepton controls, including bounded massive muons; complete components and scale/parameter changes.
CMS	Shared exact continuation and source-semantic controls; independent finite UV/branch conventions and an AP master boundary remain open.
Correlations / mapping	Bounded directed/spin controls and exact species symmetry; general composition and mapping transport remain open.
Public delivery	A genuine development install passes the public tree smoke; loop artifacts, commands and SDK/UMAMI integration remain open.
Performance	Current development timings only; distinct-point matched provider gates and required high multiplicity remain open.

Required multiplicity and acceptance

The target includes selected $2 \to 5$ processes, with $2 \to 6$ as a stretch goal. The common-provider set includes gluon jets, heavy-quark plus jets, lepton-pair plus jets, and heavy-quark/lepton-pair production with a jet. Five-final-parton gluon channels also require MadLoop/Recola controls; an unavailable public OpenLoops channel is not invented for comparison.

The generic seven-propagator reducer extension remains an external prerequisite. Rank above the supported $n + 1$ domain must reject. Required progress also includes precision-independent cold plans, persistent artifacts, optional FFT, complete correlation/mapping transport and broader certified model coverage. These requirements remain visible even when a smaller control is fast.

Independent evidence and reproducibility

The accompanying `data/measurements.json` contains the actual five timing samples, revision, process counts, cold times and checksums behind the headline tables. The frozen point fixtures and native reference drivers live in `tools/loop_reference/`. The measured driver calls the complete internal `evaluate_ordered_native` path; checks and decimal result conversion occur after the event timer. Test durations and earlier bare-only profiles are not event timings in this report.

Provider controls use MadLoop 3.7.0, Recola2 2.3.0 with its matched SM/Collier model, and OpenLoops 2.1.5. Their independent component tests establish the bounded normalization comparisons; only the fresh matched-host control timings enter the current performance table, with their input and rescue asymmetries declared. This report follows the visual structure of the NinjaRS benchmark while measuring a different object: a complete physical interference, not a single diagram and helicity.

Further reading: repository `docs/development/LOOP_MODEL_IR.md`, `docs/development/LOOP_DIRECT_R2.md`, `docs/loop-flavor-symmetry.md`, and `tools/loop_reference/README.md`; OpenLoops on-the-fly reduction · Recola · four-dimensional formulation · MadLoop standalone convention.

Living document. Update the source, measured data and compiled PDF together after each implementation milestone. Keep incomplete rows and capability limits explicit. The completion criterion is the full physics, feature, precision, packaging and per-process performance contract—not the existence of this first report.